

Aditya Singh

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LeetCode | CodeChef | Codeforces

Professional Summary

Versatile Software Engineer bridging the gap between advanced algorithmic problem-solving and scalable digital engineering. Possesses a strong foundation in complex Data Structures and Algorithms (C++) with proven global coding contest performance (CodeChef Rank 582). Equally adept at modern full-stack development, engineering real-time MERN-stack applications, integrating third-party APIs, and designing event-driven architectures. Passionate about writing highly optimized code and building robust, enterprise-grade digital solutions.

Core Skills

- **Algorithmic & Core Programming (SP Focus):** C, C++, Advanced DSA, Dynamic Programming, Time-Complexity Optimization, Object-Oriented Programming (OOP)
- **Digital & Full-Stack Technologies (DSE Focus):** JavaScript, TypeScript, React.js, Node.js, Express.js, WebSockets
- **Databases & Cloud:** MongoDB, PostgreSQL, Redis, RESTful APIs, Cloudinary, JWT
- **Tools & Concepts:** Git, GitHub, Postman, System Design, RESTful Architecture, Database Management Systems (DBMS), Operating Systems, Agile Methodologies

Work Experience

Software Development Engineer Intern | *Yuga Yatra Retail.com (Completion Certificate)* Nov 2025 – Jan 2026

- Built and optimized RESTful APIs (Node.js/Express) with secure JWT authentication, reducing backend latency by 20% and improving system throughput.
- Designed scalable MongoDB schemas and optimized database queries to efficiently process high-volume retail data.
- Integrated Cloudinary for streamlined cloud media management and developed responsive frontend components utilizing React.js.

Projects

Real-Time Auction Bidding Platform (Digital Specialist Focus) | *GitHub*

React 19, TypeScript, Node.js, Express, PostgreSQL, Prisma, Redis, BullMQ, Socket.IO, Razorpay

- Engineered an atomic transaction pipeline using Redis Lua scripting and PostgreSQL optimistic locking to eliminate race conditions during high-concurrency events.
- Automated event-driven auction lifecycles utilizing BullMQ workers and scheduled background jobs for reliable state transitions.
- Built an ACID-compliant, stake-based wallet system and integrated Razorpay with HMAC verification for secure, idempotent financial settlements.
- Delivered horizontally scalable live auction updates via Socket.IO and Redis, complete with instantaneous notifications.

Key Achievements

- **CodeChef Global Rank 582:** Achieved out of 20,000+ participants in the Starters 167 contest.
- **Technical Quiz Winner:** Secured 2nd Prize at the GNIT TechFest Technical Quiz competition.

Education

B.Tech in Computer Science

B. P. Poddar Institute of Management & Technology

Expected Graduation: 2027

CGPA: 7.97 (till 5th Sem)

Higher Secondary (Class 12)

Secondary (Class 10)

Passing Year: 2022 | *Score: 80.4%*

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